Facilities, Equipment, & Other Resources

The proposed National Indigenous Health Data Pathways Infrastructure (NIHDPI) project will be supported through coordinated facilities and resources across the Association of American Indian Physicians (AAIP), the Indigenous Health, Education, and Resources Taskforce (IHEART), and St Germaine Data Innovations, Inc. (SGDI). Collectively, these organizations provide the administrative, technical, and programmatic environment necessary to deploy, operate, and sustain the national-scale cyberinfrastructure described in this proposal.

AAIP Facilities and Organizational Resources:
AAIP will serve as the administrative and national coordination hub for the project. AAIP provides an existing organizational structure that includes national convening capabilities, established relationships with Tribal governments, academic institutions, and federal partners, and a staff team capable of supporting user engagement, training, outreach, and program coordination. AAIP's communications and operations teams will support dissemination, onboarding, resource documentation, and national meetings associated with NIHDPI. AAIP will also provide organizational access to the AAIP Board, which will serve as the project's Advisory Committee. AAIP's existing office infrastructure—including secure document management systems, communications platforms, and remote collaboration environments—will be used to support administrative and program-management activities.

IHEART Facilities and Programmatic Resources:
IHEART provides regionally distributed networks that connect K–12 partners, Tribal colleges and universities (TCUs), academic medical centers, and community healthcare organizations. These networks form the foundation for recruiting user institutions, identifying pathway programs, hosting training activities, and coordinating regional implementation pilots. IHEART will supply access to its regional leadership teams, existing educational pathway maps, and partnership structures that will be integrated into the NIHDPI infrastructure. IHEART's national reach, combined with AAIP oversight, enables broad access to Tribal-serving and Indigenous-focused health education programs, ensuring that the project's systems and services can be deployed at scale.

SGDI Technical Facilities and Computing Resources:
SGDI will provide the technical facilities needed to design, build, and maintain the project's core cyberinfrastructure. SGDI maintains secure development environments, software engineering platforms, MLOps pipelines, version-controlled repositories, and testing environments necessary for full-stack data system development. SGDI's facility in Minnesota will host the primary on-premises hardware environment supporting NIHDPI's high-performance storage, data-processing clusters, federated data-access services, and AI model training infrastructure. The facility includes climate-controlled server space, high-availability power systems, fire suppression measures, and secure access controls. SGDI will also provide continuous monitoring, logging, redundancy configurations, and operational support for all deployed systems.

Cyberinfrastructure and Software Resources:
The project will deploy hardware and systems that support data ingestion, metadata management,

storage, federated access, AI model development, and public-facing analytical dashboards. SGDI will provide software development frameworks including Python, containerized environments, automated CI/CD pipelines, infrastructure-as-code systems, and API development toolkits. Open-source software libraries, documentation platforms, and model-versioning systems will be used to ensure transparency and reproducibility. These resources will support all development, deployment, and operational phases of NIHDPI.

Communications, Training, and User-Support Resources:
AAIP and IHEART will jointly provide communication channels, virtual learning environments, national webinars, and in-person convening spaces for training and dissemination. These resources include virtual conferencing platforms, learning-management systems, secure collaboration tools, and existing online portals that will be incorporated into the NIHDPI user-support workflow. AAIP and IHEART will ensure that users across regions receive consistent onboarding, documented guidance, and help-desk support aligned with national-scale expectations.

Institutional Partnerships and External Access:
NIHDPI will integrate with a wide array of external institutions, including TCUs, Tribal education departments, K–12 pathway programs, health professional schools, and Tribal Health/IHS clinical sites. Each partner institution will contribute its own local computing, security, and data-management environments through federated connectors, rather than centralized data transfer. These resources will enable secure, institution-controlled read access to restricted data, ensuring compliance with Indigenous data sovereignty requirements while supporting national analytics.

Human Resources and Expertise:
The project team includes senior leadership from AAIP, IHEART, and SGDI with combined expertise in Indigenous health, pathway program development, data science, cyberinfrastructure deployment, software engineering, AI tooling, and project management. SGDI's technical team includes data engineers, data scientists, MLOps engineers, software consultants, evaluators, and administrative support required to operate and maintain the infrastructure. AAIP and IHEART provide medical, educational, and organizational expertise essential for aligning the infrastructure with community needs, scientific priorities, and workforce-development goals.

Collectively, these facilities, technical capabilities, organizational structures, and partnership networks ensure that NIHDPI has the operational foundation and institutional support needed to develop, deploy, and sustain national-scale data cyberinfrastructure.